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### Light emitting element

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#### Abstract

A light emitting element comprises a first n-side semiconductor layer, a first active layer having a first well layer containing indium, a first p-side semiconductor layer, a second n-side semiconductor layer in contact with the first p-side semiconductor layer, a second active layer having a second well layer containing indium, and a second p-side semiconductor, each formed of a nitride semiconductor. The second active layer has a first intermediate layer positioned closer to the first active layer than is the second well layer and containing indium. An indium composition ratio of the first well layer is less than an indium composition ratio of the second well layer. An indium composition ratio of the first intermediate layer is less than an indium composition ratio of the first well layer. A thickness of the first intermediate layer is less than a thickness of the second well layer.

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## Background/Summary

### CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2022-022046, filed on Feb. 16, 2022, and Japanese Patent Application No. 2022-132555, filed on Aug. 23, 2022, the disclosures of which are hereby incorporated by reference in their entireties.

### BACKGROUND

(2) The present disclosure relates to a light emitting element. Japanese Translation of PCT International Application Publication No. JP-T-2009-527920 discloses a light emitting element including a tunnel junction and having a plurality of active regions, each emitting light of different wavelengths.

(3) Semiconductor layers having high impurity concentrations are needed for the p-type layer and the n-type layer that form a tunnel junction. Diffusion of p-type impurity from a tunnel-junction-forming p-type layer towards the second active layer side is believed to be one of the causes of a reduced output of a light emitting element having a tunnel junction.

(4) An object of the present disclosure is to provide a high output light emitting element.

(5) According to an embodiment of the present disclosure, a light emitting element comprises, successively, a first n-side semiconductor layer, a first active layer having a first well layer containing indium, a first p-side semiconductor layer, a second n-side semiconductor layer in contact with the first p-side semiconductor layer, a second active layer having a second well layer containing indium, and a second p-side semiconductor, each being formed of a nitride semiconductor, wherein the second active layer has a first intermediate layer positioned closer to the first active layer than the second well layer is and containing indium, the first well layer has a lower indium composition ratio than the indium composition ratio of the second well layer, the first intermediate layer has a lower indium composition ratio than the indium composition ratio of the first well layer, and the first intermediate layer is less in thickness than the second well layer.

(6) According to the present disclosure, a high output light emitting element can be provided.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic cross-sectional view of a light emitting element according to a first embodiment.

(2) FIG. 2 is a schematic cross-sectional view explaining a first superlattice layer of the light emitting element according to the first embodiment.

(3) FIG. 3 is a schematic cross-sectional view explaining a first active layer of the light emitting element according to the first embodiment.

(4) FIG. 4 is a schematic cross-sectional view explaining a second superlattice layer of the light emitting element according to the first embodiment.

(5) FIG. 5 is a schematic cross-sectional view explaining a second active layer of the light emitting element according to the first embodiment.

(6) FIG. 6A is a graph showing the output measurement results of the first wavelength light emitted by the first active layers of the light emitting element samples.

(7) FIG. 6B is a graph showing the output measurement results of the second wavelength light emitted by the first active layers of the light emitting element samples.

(8) FIG. 7 is a schematic cross-sectional view of a light emitting element according to a second embodiment.

(9) FIG. 8 is a schematic cross-sectional view explaining a third active later of the light emitting

element according to the second embodiment.

## DETAILED DESCRIPTION

(10) Certain embodiments will be explained with reference to the accompanying drawings. In the drawings, the same reference numerals are used to denote the same elements. The drawings are schematic representations of the embodiments.

(11) Therefore, the scale, spacing, or positional relationships of the members might be exaggerated, or a certain portion of a member omitted. An end face view showing only a cut section might be used as a cross-sectional view.

(12) In the description below, the constituent elements having practically the same functions are denoted by common reference numerals for which explanation may be omitted. Terms indicating specific directions or positions (e.g., “upper,” “on,” “lower,” “under,” and other terms including or related to these) may be used. These terms, however, are merely used in order to make the relative directions or positions in the drawings being referenced more easily understood. As long as the relationship between relative directions or positions indicated with the terms such as “upper,” “on,” “lower,” “under,” or the like is the same as those in a referenced drawing, the layout of the elements in other drawings or actual products outside of the present disclosure does not have to be the same as those shown in the referenced drawing. The positional relationship expressed with the term “on” in the present specification includes the case in which a member is in contact with another member, and the case in which a member is not in contact with, but positioned above another member.

### First Embodiment

(13) As shown in FIG. 1, a light emitting element **1** according to a first embodiment includes a substrate **100**, a semiconductor structure body **10**, a p-side electrode **11**, and an n-side electrode **12**.

(14) A substrate **100** supports a semiconductor structure body **10**. For the material for the substrate **100**, for example, sapphire, silicon, gallium nitride, or the like can be used. In the case of using a sapphire substrate as the substrate **100**, the semiconductor structure body **10** is disposed, for example, on C-plane of the sapphire substrate.

(15) A semiconductor structure body **10** includes, successively from the substrate **100** side, a first n-side semiconductor layer **20**, a first active layer **40**, a first p-side semiconductor layer **50**, a second n-side semiconductor layer **60**, a second active layer **90**, and a second p-side semiconductor layer **110**. The semiconductor structure body **10** can further include a first superlattice layer **30** disposed between the first n-side semiconductor layer **20** and the first active layer **40**.

(16) The first n-side semiconductor layer **20**, the first superlattice layer **30**, the first active layer **40**, the first p-side semiconductor layer **50**, the second n-side semiconductor layer **60**, the second active layer **90**, and the second p-side semiconductor layer **110** are individually formed of a nitride semiconductor. The nitride semiconductor can be any semiconductor obtained by varying the composition ratio  $x$  and  $y$  within their ranges in the chemical formula of  $\text{In}_{0.5}\text{Al}_{0.5}\text{Ga}_{0.5}\text{In}_{1-x-y}\text{N}$  ( $0 \leq x \leq 1$ ,  $0 \leq y \leq 1$ ,  $x+y \leq 1$ ). For example, the semiconductor structure body **10** can be epitaxially grown on the substrate **100**.

(17) The first n-side semiconductor layer **20** has an n-type layer containing an n-type impurity. The n-type layer contains, for example, silicon (Si) as the n-type impurity. The n-type layer may contain germanium (Ge) as the n-type impurity. The first n-side semiconductor layer **20** has only to have the function of supplying electrons, and may include an undoped layer formed without being intentionally doped with an n-type impurity or a p-type impurity. In the case in which the undoped layer is adjacent to a layer intentionally doped with an n-type impurity and/or a p-type impurity, the undoped layer might contain the n-type impurity and/or the p-type impurity diffused from the adjacent layer.

(18) The first p-side semiconductor layer **50** and the second p-side semiconductor layer **110** each have a p-type layer containing a p-type impurity. The p-type layer contains magnesium (Mg), for example, as the p-type impurity. The first p-side semiconductor layer **50** and the second p-side

semiconductor layer **110** have only to have the function of supplying positive holes, and may include an undoped layer.

(19) The first active layer **40** and the second active layer **90** have a multi-quantum well structure including a plurality of well layers and a plurality of barrier layers as described later. The peak wavelength of the light emitted by the first active layer **90** is shorter than the peak wavelength of the light emitted by the second active layer **90**. For example, the first active layer **40** emits blue light and the second active layer **90** emits green light. The peak emission wavelength of blue light is in a range of 430 nm to 490 nm. The peak emission wavelength of green light is in a range of 500 nm to 540 nm. The first active layer **40** may emit ultraviolet light. The peak emission wavelength of ultraviolet light is 410 nm or lower.

(20) The second n-side semiconductor layer **60** is in contact with the first p-side semiconductor layer **50**. The second n-side semiconductor layer **60** has a first tunnel junction layer **70** disposed in contact with the first p-side semiconductor layer **50**. The first tunnel junction layer **70** in contact with the first p-side semiconductor layer **50** forms a tunnel junction with the first p-side semiconductor layer **50**. The second n-side semiconductor layer **60** can further have a second superlattice layer **80** disposed between the first tunnel junction layer **70** and the second active layer **90**. The first tunnel junction layer **70** includes an n-type layer having a higher n-type impurity concentration than the n-type impurity concentration of the second superlattice layer **80**. The first tunnel junction layer **70** includes an n-type layer having an n-type impurity concentration, for example, of  $1 \times 10^{20} \text{ cm}^{-3}$  to  $5 \times 10^{21} \text{ cm}^{-3}$ .

(21) Providing a first superlattice layer **30** and a second superlattice layer **80** can reduce lattice mismatch between the substrate **100** and the semiconductor structure body **10**, thereby reducing crystal defects in the semiconductor structure body **10**.

(22) The first n-side semiconductor layer **20** has an n-side contact face **20a** on which no semiconductor layer is disposed. An n-side electrode **12** is disposed on the n-side contact face **20a**. The n-side electrode **12** is electrically connected to the first n-side semiconductor layer **20**.

(23) A p-side electrode **11** is disposed on the upper face of the second p-side semiconductor layer **110**. The p-side electrode **11** is electrically connected to the second p-side semiconductor layer **110**.

(24) A forward voltage is applied across the p-side electrode **11** and the n-side electrode **12**. At this time, a forward current is supplied across the first active layer **40** and the second active layer **90** allowing the first active layer **40** and the second active layer **90** to emit light.

(25) Providing a second active layer **90** on a first active layer **40** can increase the per unit area output as compared to a light emitting element having a single active layer. The light emitting element **1** outputs mixed color light of first wavelength light (e.g., blue light) emitted by the first active layer **40** and second wavelength light (e.g., green light) emitted by the second active layer **90**.

(26) When a forward voltage is applied across the p-side electrode **11** and the n-side electrode **12**, a reverse voltage would apply across the second n-side semiconductor layer **60** and the first p-side semiconductor layer **50**. The n-type layer and the p-type layer in the second n-side semiconductor layer **60** and the first p-side semiconductor layer **50** that form a tunnel junction having high impurity concentrations can reduce the width of the depletion layer formed by the junction between the second n-side semiconductor layer **60** and the first p-side semiconductor layer **50**. This allows for the tunneling of electrons present in the valence band of the p-type layer to the conduction band of the n-type layer, thereby facilitating the current flow across the second n-side semiconductor layer **60** and the first p-side semiconductor layer **50**.

(27) The first superlattice layer **30**, the first active layer **40**, the second superlattice layer **80**, and the second active layer **90** will be explained in detail below.

(28) First Superlattice Layer

(29) As shown in FIG. 2, a first superlattice layer **30** has a plurality of third layers **31** and a plurality of fourth layers **32**. The first superlattice layer **30** can have from 15 to 25 pairs of a third

layer **31** and a fourth layer **32**. The first superlattice layer **30** can have, for example, 20 third layers **31** and 20 fourth layers **32**. In the first superlattice layer **30**, a fourth layer **32** is positioned as the lowermost layer, and a third layer **31** is positioned as the uppermost layer. The fourth layers **32** and the third layers **31** are alternately disposed from the fourth layer **32** positioned as the lowermost layer to the third layer **31** positioned as the uppermost layer.

(30) The fourth layers **32** include a fourth layer **32** positioned between two adjacent third layers **31** among the third layers **31**. Furthermore, a fourth layer **32** is also positioned between the lowermost third layer **31** and the first n-side semiconductor layer **20**.

(31) The composition of the third layers **31** is different from the composition of the fourth layers **32**. The third layers **31** contain indium (In). The third layers **31** are formed of undoped indium gallium nitride (InGaN), for example. The indium composition ratio of the third layers **31** can be set, for example, in a range of 3% to 10%. The fourth layers **32** are formed of undoped gallium nitride (GaN), for example. The n-type impurity concentration of the third layers **31** and the fourth layers **32** can be set, for example, in a range of  $1 \times 10^{17}/\text{cm}^3$  to  $1 \times 10^{19}/\text{cm}^3$ . The n-type impurity concentration of the third layers **31** and the fourth layers **32** refers to the highest n-type impurity concentration in the third layers **31** and the fourth layers **32**.

(32) The thickness of the individual third layers **31** is less than the thickness of the fourth layers **32**. The thickness of the individual third layers **31** can be, for example, in a range of 0.5 nm to 1.5 nm. The thickness of the individual fourth layers **32** can be, for example, in a range of 1.5 nm to 3 nm.

(33) First Active Layer

(34) As shown in FIG. 3, a first active layer **40** has a plurality of first well layers **41** and at least one first barrier layer **42**. The first active layer **40** can have, for example, 7 first well layers **41** and 6 first barrier layers **42**. The individual first barrier layer **42** is positioned between two adjacent first well layers **41** among the first well layers **41**. The first active layer **40** can further have a third barrier layer **43** positioned as the lowermost layer and a fourth barrier layer **44** positioned as the uppermost layer in the first active layer **40**. A first well layer **41** is provided between the lowermost first barrier layer **42** among the first barrier layers **42** and the third barrier layer **43**. A first well layer **41** is provided between the uppermost first barrier layer **42** among the first barrier layers **42** and the fourth barrier layer **44**. The first well layers **41** and the first barrier layers **42** are alternately formed between the third barrier layer **43** and the fourth barrier layer **44**.

(35) The first well layers **41** contain indium. The first well layers **41** are formed, for example, of undoped indium gallium nitride. The indium composition ratio of the first well layers **41** can be set, for example, in a range of 12% to 18%.

(36) The first barrier layers **42**, the third barrier layer **43**, and the fourth barrier layer **44** individually have a band gap larger than the band gap of a first well layer **41**. The first barrier layers **42**, the third barrier layer **43**, and the fourth barrier layer **44** are formed, for example, of gallium nitride.

(37) The thickness of respective of the first barrier layers **42** and the thickness of the fourth barrier layer **44** are larger than the thickness of the first well layers **41**. The thickness of the respective first well layers can be set, for example, in a range of 2.5 nm to 5 nm. The thickness of respective of the first barrier layers **42**, the third barrier layer **43**, and the fourth barrier layer **44** can be set, for example, in a range of 3 nm to 6 nm.

(38) Second Superlattice Layer

(39) As shown in FIG. 4, a second superlattice layer **80** has a plurality of first layers **81** and a plurality of second layers **82**. The second superlattice layer **80** can have, for example, in a range of 15 to 25 pairs of a first layer **81** and a second layer **82**. The second superlattice layer **80** can have, for example, 20 first layers **81** and 20 second layers **82**. In the second superlattice layer **80**, a second layer **82** is positioned as the lowermost layer, and a first layer **81** is positioned as the uppermost layer. The second layers **82** and the first layers **81** are alternately formed from the second layer **82** positioned as the lowermost layer to the first layer **81** positioned as the uppermost

layer.

(40) The second layers **82** include a second layer **82** which is positioned between two adjacent first layers **81** among the first layers **81**. A second layer **82** is also provided between the lowermost first layer **81** and the first tunnel junction layer **70**.

(41) The composition of the first layers **81** differ from the composition of the second layers **82**. The first layers **81** contain indium. The first layers **81** are made, for example, of silicon-doped indium gallium nitride. The indium composition ratio of the first layers **81** can be set, for example, in a range of 5% to 10%. The second layers **82** are made, for example, of silicon-doped gallium nitride. The n-type impurity concentration of the first layers **81** and the second layers **82** can be set, for example, in a range of  $1 \times 10^{17} / \text{cm}^3$  to  $1 \times 10^{20} / \text{cm}^3$ . The n-type impurity concentration of the first layers **81** refers to the highest n-type impurity concentration in the first layers **81**. The n-type impurity concentration of the second layers **82** refers to the highest n-type impurity concentration in the second layers **82**.

(42) The thickness of the individual first layers **81** is less than the thickness of the second layers **82**. The thickness of the first layers **81** can be set, for example, in a range of 0.5 nm to 1.5 nm. The thickness of the second layers **82** can be set, for example, in a range of 1.5 nm to 3 nm.

(43) Second Active Layer

(44) As shown in FIG. 5, a second active layer **90** has a second well layer **91**, a first intermediate layer **93**, a second barrier layer **92**, a fifth barrier layer **95**, and a sixth barrier layer **96**.

(45) FIG. 5 shows an example in which the second active layer **90** has a single second well layer **91** and a single second barrier layer **92**. The second active layer **90** may have multiple pairs of a second well layer **91** and a second barrier layer **92**. In this case, individual ones of the second barrier layers **92** is positioned between two adjacent second well layers **91** among the second well layers **91**.

(46) The first intermediate layer **93** is positioned closer to the first active layer **40** than the second well layer **91** is. The second barrier layer **92** is positioned between the second well layer **91** and the first intermediate layer **93**. In the second active layer **90**, the fifth barrier layer **95** is positioned as the lowermost layer, and the sixth barrier layer **96** is positioned as the uppermost layer. In the case in which multiple pairs of a second well layer **91** and a second barrier layer **92** are provided, the first intermediate layer **93** is positioned between the lowermost second barrier layer **92** among the second barrier layers **92** and the fifth barrier layer **95**. The second well layer **91** is positioned between the second barrier layer **92** and the sixth barrier layer **96**. In the case in which multiple pairs of a second well layer **91** and a second barrier layer **92** are provided, the uppermost second well layer **91** among the second well layers **91** is positioned between the uppermost second barrier layer **92** among the second barrier layers **92** and the sixth barrier layer **96**.

(47) The second well layer **91** contains indium. The second well layer **91** is formed, for example, of undoped indium gallium nitride. The indium composition ratio of the second well layer **91** is higher than the indium composition ratio of the first well layers **41** of the first active layer **40**. The indium composition ratio of the second well layer **91** can be set, for example, in a range of 20% to 25%.

(48) The first intermediate layer **93** contains indium. The first intermediate layer **93** is formed, for example, of undoped indium gallium nitride. The indium composition ratio of the first intermediate layer **93** is lower than the indium composition ratio of the first well layers **41** of the first active layer **40**. The indium composition ratio of the first intermediate layer **93** is higher than the indium composition ratio of the first layers **81** of the second superlattice layer **80**. The indium composition ratio of the first intermediate layer **93** can be set, for example, in a range of 8% to 12%.

(49) The second barrier layer **92**, the fifth barrier layer **95**, and the sixth barrier layer **96** individually have a band gap larger than the band gaps of the second well layer **91** and the first intermediate layer **93**. The second barrier layer **92**, the fifth barrier layer **95**, and the sixth barrier layer **96** are formed, for example, of gallium nitride.

(50) The thickness of the first intermediate layer **93** is less than the thickness of the second well

layer **91**. The thickness of the first intermediate layer **93** can be set, for example, in a range of 1 nm to 2 nm. The thickness of the second well layer **91** is less than the thickness of the first well layers **41**. The thickness of the second well layer **91** can be set, for example, in a range of 2 nm to 3 nm. (51) The thicknesses of the second barrier layer **92**, the fifth barrier layer **95**, and the six barrier layer **96** are all larger than the thickness of the second well layer **91**. The thickness of individual ones of the second barrier layer **92**, the fifth barrier layer **95**, and the six barrier layer **96** can be set, for example, in a range of 3 nm to 15 nm.

(52) According to this embodiment, positioning a first intermediate layer **93** between the second well layer **91** and the tunnel junction formed by the first p-side semiconductor layer **50** and the second n-side semiconductor layer **60** can increase the distance between the second well layer **91** and the tunnel-junction-forming p-type layer, thereby making it difficult for the p-type impurity (e.g., magnesium) to diffuse from the tunnel-junction-forming p-type layer to the second well layer **91**. This can reduce the deterioration of the crystalline quality of the second well layer **91** attributable to the diffusion of p-type impurity thereto, thereby increasing the light output of the second well layer **91**.

(53) The distance between the second well layer **91** and the tunnel junction can possibly be increased by increasing the thickness of the barrier layer disposed between the second well layer **91** and the second n-side semiconductor layer **60** without disposing a first intermediate layer **93** in the second active layer **90**. In this case, the barrier layer having a band gap larger than the second well layer **91** can become a barrier against the movement of electrons and positive holes to readily increase the forward voltage.

(54) The first intermediate layer **93** having a band gap smaller than the band gaps of the second barrier layer **92** and the fifth barrier layer **95** disposed in contact with the first intermediate layer **93**, as well as a thickness less than that of the second well layer **91**, can achieve a tunnelling effect to reduce the forward voltage. The thickness of the first intermediate layer **93** is preferably set, for example, in a range of 1 nm to 2 nm.

(55) The indium composition ratio of the first intermediate layer **93** being lower than the indium composition ratio of the first well layers **41** of the first active layer **40** can reduce the absorption of the light emitted by the first well layers **41** by the first intermediate layer **93**. The indium composition ratio of the first intermediate layer **93** is preferably set, for example, in a range of 8% to 12%.

(56) In order to further reduce the absorption of the light from the first well layer **41** by the second active layer **90**, the quantity of second well layers **91** is preferably set to be less than the quantity of the first well layers **41**.

(57) FIG. 6A is a graph showing the measurement results of the first wavelength light output by the first active layer **40** in individual ones of the light emitting element samples tested. The first wavelength light is blue light.

(58) Samples were prepared by varying the quantity of second well layers **91** in the second active layer **90** which included no first intermediate layer **93** to be one layer, two layers, and three layers, and the first wavelength light output from the first active layer **40** of individual sample was measured. The measurement results are indicated by filled circles.

(59) A sample in which the second active layer **90** included 1 second well layer **91** and 1 first intermediate layer **93** was prepared, and the first wavelength light output from the first active layer **40** was measured. The measurement result is indicated by the open circle.

(60) The first wavelength light outputs shown along the vertical axis in FIG. 6A are relative values when compared to the first wavelength light output of the sample which included no first intermediate layer **93** and 3 second well layers **91** assumed as 100%.

(61) FIG. 6B is a graph showing the measurement results of the second wavelength light output from the second active layer **90** in individual ones of the light emitting element samples tested. The second wavelength light is green light.



(62) Samples were prepared by varying the quantity of second well layers **91** in the second active layer **90** which included no first intermediate layer **93** to be one layer, two layers, and three layers, and the second wavelength light output from the second active layer **90** of individual sample was measured. The measurement results are indicated by filled circles.

(63) A sample in which the second active layer **90** included 1 second well layer **91** and 1 first intermediate layer **93** was prepared, and the second wavelength light output from the second active layer **90** was measured. The measurement result is indicated by the open circle.

(64) The second wavelength light outputs shown along the vertical axis in FIG. 6B are relative values when compared to the second wavelength light output of the sample which included no first intermediate layer **93** and 3 second well layers **91** assumed as 100%.

(65) Individual ones of the samples used in the measurements described above had the constituents described below.

(66) The substrate **100** was a sapphire substrate.

(67) The first n-side semiconductor layer **20** contained silicon as a n-type impurity. The silicon concentration of the first n-side semiconductor layer **20** was about  $1 \times 10^{19}$  cm<sup>-3</sup>. The silicon concentration of the first n-side semiconductor layer **20** refers to the highest silicon concentration in the first n-side semiconductor layer **20**. The thickness of the first n-side semiconductor layer **20** was about 10  $\mu$ m.

(68) The first superlattice layer **30** had 20 third layers **31** and 20 fourth layers **32**. The third layers **31** were undoped indium gallium nitride layers. The indium composition ratio of the third layers **31** was about 7%. The thickness of the third layers **31** were about 1 nm. The fourth layers **32** were undoped gallium nitride layers. The thickness of the fourth layers **32** were about 2 nm.

(69) The first active layer **40** had 7 first well layers **41**. The first well layers **41** were undoped indium gallium nitride layers. The indium composition ratio of the first well layers **41** was about 15%. The thickness of the first well layers **41** were about 3.4 nm.

(70) The first p-side semiconductor layer **50** contained magnesium as a p-type impurity. The magnesium concentration of the first p-side semiconductor **50** was about  $3 \times 10^{20}$  cm<sup>-3</sup>. The magnesium concentration of the first p-side semiconductor **50** refers to the highest magnesium concentration in the first p-side semiconductor **50**. The thickness of the first p-side semiconductor **50** was about 140 nm.

(71) The first tunnel junction layer **70** included a silicon-doped n-type gallium nitride layer. The silicon concentration of the first tunnel junction layer **70** was about  $8 \times 10^{20}$  cm<sup>-3</sup>. The silicon concentration of the first tunnel junction layer **70** refers to the highest silicon concentration in the first tunnel junction layer **70**. The thickness of the first tunnel junction layer **70** was about 2 nm.

(72) The second superlattice layer **80** had 20 first layers **81** and 20 second layers **82**. The first layers **81** were silicon-doped indium gallium nitride layers. The indium composition ratio of the first layers **81** was about 7%. The thickness of the individual first layers **81** were about 1 nm. The second layers **82** were silicon-doped gallium nitride layers. The thickness of the individual second layers **82** were about 2 nm. The silicon concentration of the first layers **81** and the second layers **82** was about  $2 \times 10^{19}$  cm<sup>-3</sup>. The silicon concentration of the first layers **81** and the second layers **82** refers to the highest silicon concentration in the first layers **81** and the second layers **82**.

(73) Individual second well layer **91** of the second active layer **90** was an undoped indium gallium nitride layer. The indium composition ratio of the individual second well layer **91** was about 23%. The thickness of the individual second well layers **91** was about 2.8 nm. In the sample in which the second active layer **90** included a first intermediate layer **93**, the first intermediate layer **93** was an undoped indium gallium nitride layer. The indium composition of the first intermediate layer **93** was about 10%. The thickness of the individual first intermediate layer **93** was about 1.5 nm.

(74) The second p-side semiconductor layer **110** contained magnesium as a p-type impurity. The magnesium concentration of the second p-side semiconductor layer **110** was about  $3 \times 10^{20}$

cm.sup.3. The magnesium concentration of the second p-side semiconductor layer **110** refers to the highest magnesium concentration in the second p-side semiconductor layer **110**. The thickness of the second p-side semiconductor layer **110** was about 115 nm.

(75) As shown in FIG. **6A**, as to the samples not including a first intermediate layer **93** (the measurement points indicated by filled circles), as the quantity of second well layers **91** decreased, the absorption of the first wavelength light (blue light) in the second well layer(s) **91** tended to decline to increase the first wavelength light output.

(76) On the other hand, as shown in FIG. **6B**, as to the samples not including a first intermediate layer **93** (the measurement points indicated by filled circles), as the quantity of second well layers **91** decreased, the second wavelength light (green light) output tended to decrease.

(77) The sample having 1 second well layer **91** and 1 first intermediate layer **93** was able to achieve a higher second wavelength light output (the open circle in FIG. **6B**) than the sample having one second layer **91** and no first intermediate layer **93**, while maintaining almost the same first wavelength light output (the open circle in FIG. **6A**) as that of the sample which had one second layer **91** and no first intermediate layer **93**. Furthermore, the sample having 1 second well layer **91** and 1 first intermediate layer **93** was able to increase the output of the light emitting element output combining the first wavelength light output and the second wavelength light output, as compared to the sample which had three second well layers **91** and no first intermediate layer **93**.

#### Second Embodiment

(78) As shown in FIG. **7**, a light emitting element **2** according to a second embodiment includes a substrate **100**, a semiconductor structure body **10**, a p-side electrode **11**, and an n-side electrode **12**. The light emitting element **2** of the second embodiment has practically the same structure as the light emitting element **1** of the first embodiment except for further including in the semiconductor structure body **10** a third n-side semiconductor layer **120**, a third active layer **150**, and a third p-side semiconductor layer **160**. The third n-side semiconductor layer **120**, the third active layer **150**, and the third p-side semiconductor layer **160** are formed of a nitride semiconductor.

(79) The third n-side semiconductor layer **120** is in contact with the second p-side semiconductor layer **110**. The third n-side semiconductor layer **120** has a second tunnel junction layer **130** disposed in contact with the second p-side semiconductor layer **110**. The second tunnel junction layer **130** in contact with the second p-side semiconductor layer **110** forms a tunnel junction with the p-side semiconductor layer **110**. The third n-side semiconductor layer **120** can further have a third superlattice layer **140** disposed between the second tunnel junction layer **130** and the third active layer **150**. The third superlattice layer **140** has a similar structure to that of the second superlattice layer **80**. Providing a third superlattice layer **140** can reduce lattice mismatch between the second tunnel junction layer **130** and the third active layer **150**, thereby reducing crystal defects in the semiconductor structure body **10**.

(80) The second tunnel junction layer **130** has a similar structure to that of the first tunnel junction layer **70**. The second tunnel junction layer **130** includes an n-type layer having an n-type impurity concentration higher than the n-type impurity concentration of the third superlattice layer **140**. The second tunnel junction layer **130** includes an n-type layer having an n-type impurity concentration, for example, in a range of  $1 \times 10^{20} / \text{cm}^3$  to  $5 \times 10^{21} / \text{cm}^3$ .

(81) The third active layer **150** is disposed on the third n-side semiconductor layer **120**. The peak wavelength of the light from the third active layer **150** is longer than the peak wavelengths of the light from the first active layer **40** and the light from the second active layer **90**. The third active layer **150** emits red light, for example. The peak emission wavelength of red light is from 615 nm to 750 nm.

(82) As shown in FIG. **8**, the third active layer **150** has a third well layer **151**, a second intermediate layer **153**, a seventh barrier layer **152**, an eighth barrier layer **155**, and a ninth barrier layer **156**. The third active layer **150** may have multiple pairs of a third well layer **151** and a seventh barrier layer **152**. In this case, individual ones of the seventh barrier layers **152** are positioned between two

adjacent third well layers **151** among the third well layers **151**.

(83) The second intermediate layer **153** is positioned closer to the second active layer **90** than the third well layer **151** is. The seventh barrier layer **152** is positioned between the third well layer **151** and the second intermediate layer **153**. In the third active layer **150**, the eighth barrier layer **155** is positioned as the lowermost layer, and the ninth barrier layer **156** is positioned as the uppermost layer. In the case of providing multiple pairs of a third well layer **151** and a seventh barrier layer **152**, the second intermediate layer **153** is positioned between the lowermost seventh barrier layer **152** among the seventh barrier layers **152** and the eighth barrier layer **155**. The third well layer **151** is positioned between the seventh barrier layer **152** and the ninth barrier layer **156**. In the case of providing multiple pairs of a third well layer **151** and a seventh barrier layer **152**, the uppermost third well layer **151** among the third well layers **151** is positioned between the uppermost seventh barrier layer **152** among the seventh barrier layers **152** and the ninth barrier layer **156**.

(84) The third well layer **151** contains indium. The third well layer **151** is formed, for example, of undoped indium gallium nitride. The indium composition ratio of the third well layer **151** is higher than the indium composition ratio of the second well layer **91**. The indium composition ratio of the third well layer **151** can be set, for example, in a range of 25% to 40%, more preferably 30% to 35%. The thickness of the third well layer **151** can be set, for example, in a range of 2 nm to 4 nm.

(85) The second intermediate layer **153** contains indium. The second intermediate layer **153** is formed, for example, of undoped indium gallium nitride. The indium composition ratio of the second intermediate layer **153** is lower than the indium composition ratio of the second well layer **91**. The indium composition ratio of the second intermediate layer **153** can be the same as or lower than the indium composition ratio of the first intermediate layer **93**. The thickness of the second intermediate layer **153** is less than the thickness of the third well layer **151**.

(86) The band gaps of the seventh barrier layer **152**, the eighth barrier layer **155**, and the ninth barrier layer **156** are larger than the band gap of the third well layer **151** and the band gap of the second intermediate layer **153**. The seventh barrier layer **152**, the eighth barrier layer **155**, and the ninth barrier layer **156** are formed, for example, of gallium nitride.

(87) The thickness of respective of the seventh barrier layer **152**, the eighth barrier layer **155**, and the ninth barrier layer **156** are larger than the thickness of the third well layer **151**. The thickness of the seventh barrier layer **152**, the thickness of the eighth barrier layer **155**, and the thickness of the ninth barrier layer **156** can be set, for example, in a range of 3 nm to 15 nm.

(88) The third p-side semiconductor layer **160** is disposed on the third active layer **150**. The third p-side semiconductor layer **160** has a p-type layer containing a p-type impurity. The p-type layer of the third p-side semiconductor layer **160** contains magnesium (Mg) as a p-type impurity, for example. The third p-side semiconductor layer **160** has only to have the function of supplying positive holes, and may include an undoped layer.

(89) A p-side electrode **11** is disposed on the upper face of the third p-side semiconductor layer **160**. The p-side electrode **11** is electrically connected to the third p-side semiconductor layer **160**.

(90) A forward voltage is applied across the p-side electrode **11** and the n-side electrode **12** in a light emitting element **2** of the second embodiment. At this time, a forward current is supplied to the first active layer **40**, the second active layer **90**, and the third active layer **150**, allowing the first active layer **40**, the second active layer **90**, and the third active layer **150** to emit light.

(91) Providing a second active layer **90** on a first active layer **40**, and a third active layer **150** on the second active layer **90** can increase the per unit area output as compared to a light emitting element having a single active layer or a light emitting element **1** of the first embodiment which has two active layers. The light emitting element **2** outputs mixed color light of first wavelength light (e.g., blue light) emitted by the first active layer **40**, second wavelength light (e.g., green light) emitted by the second active layer **90**, and third wavelength light (e.g., red light) emitted by the third active layer **150**.

(92) When a forward voltage is applied across the p-side electrode **11** and the n-side electrode **12**, a

reverse voltage would apply across the second n-side semiconductor layer **60** and the first p-side semiconductor layer **50** as well as across the third n-side semiconductor layer **120** and the second p-side semiconductor layer **110**. The n-type layer and the p-type layer in the second n-side semiconductor layer **60** and the first p-side semiconductor layer **50** that form a tunnel junction having high impurity concentrations, as well as the n-type layer and the p-type layer in the third n-side semiconductor layer **120** and the second p-side semiconductor layer **110** that form a tunnel junction having high impurity concentrations, can reduce the width of the depletion layer formed by the junction between the second n-side semiconductor layer **60** and the first p-side semiconductor layer **50**, and the width of the depletion layer formed by the junction between the third n-side semiconductor layer **120** and the second p-side semiconductor layer **110**. This allows for the tunneling of electrons present in the valence band of individual ones of the p-type layers to the conduction band of individual ones of the n-type layers, thereby facilitating the current flow across the second n-side semiconductor layer **60** and the first p-side semiconductor layer **50** and across the third n-side semiconductor layer **120** and the second p-side semiconductor layer **110**.

(93) According to the second embodiment, positioning a second intermediate layer **153** between the third well layer **151** and the tunnel junction formed by the second p-side semiconductor layer **110** and the third n-side semiconductor layer **120** can increase the distance between the third well layer **151** and the tunnel-junction-forming p-type layer, thereby making it difficult for the p-type impurity (e.g., magnesium) to diffuse from the tunnel-junction-forming p-type layer to the third well layer **151**. This can reduce the deterioration of the crystalline quality of the third well layer **151** attributable to the diffusion of p-type impurity thereto, thereby increasing the light output of the third well layer **151**.

(94) The second intermediate layer **153** having a band gap smaller than the band gaps of the seventh barrier layer **152** and the eighth barrier layer **155** disposed in contact with the second intermediate layer **153** as well as a thickness less than that of the third well layer **151** can achieve a tunneling effect to reduce the forward voltage. The thickness of the second intermediate layer **153** is preferably set, for example, in a range of 1 nm to 2 nm. The thickness of the second intermediate layer **153** can be set as equal to or less than the thickness of the first intermediate layer **93**.

(95) The indium composition ratio of the second intermediate layer **153** being lower than the indium composition ratio of the second well layer **91** can reduce the absorption of the light emitted from the second well layer **91** by the second intermediate layer **153**. The indium composition ratio of the second intermediate layer **153** is preferably set, for example, in a range of 8% to 12%.

(96) In order to further reduce the absorption of the light emitted from the first well layers **41** by the third active layer **150**, the quantity of third well layers **151** is preferably set to be less than the quantity of first well layers **41**. The quantity of third well layers **151** can be set, for example, as one.

(97) The first active layer **40** and the second active layer **90** of a light emitting element **1** according to the first embodiment may be individually ON/OFF controlled. For example, the first active layer **40** and the second active layer **90** may be individually ON/OFF controlled by providing an electrode electrically connected to the second n-side semiconductor layer **60**. The first active layer **40**, the second active layer **90**, and the third active layer **150** of a light emitting element **2** according to the second embodiment may be individually ON/OFF controlled. For example, the first active layer **40**, the second active layer **90**, and the third active layer **150** may be individually ON/OFF controlled by providing an electrode electrically connected to the second n-side semiconductor layer **60** and an electrode electrically connected to the third n-side semiconductor layer **120**.

(98) In the foregoing, certain embodiments of the present invention have been explained with reference to specific examples. The present invention, however, is not limited to these specific examples. All forms implementable by a person skilled in the art by suitably making design changes based on any of the embodiments of the present invention described above also fall within the scope of the present invention so long as they encompass the subject matter of the present

invention. Furthermore, various modifications and alterations within the spirit of the present invention that could have been made by a person skilled in the art also fall within the scope of the present invention.

(99) Embodiments of the present invention includes any of the light emitting elements described below.

(100) [Item 1] A light emitting element comprising, successively: a first n-side semiconductor layer, a first active layer having a first well layer containing indium, a first p-side semiconductor layer, a second n-side semiconductor layer in contact with the first p-side semiconductor layer, a second active layer having a second well layer containing indium, and a second p-side semiconductor, each being formed of a nitride semiconductor, wherein the second active layer has a first intermediate layer positioned closer to the first active layer than the second well layer is and containing indium, the first well layer has a lower indium composition ratio than the indium composition ratio of the second well layer, the first intermediate layer has a lower indium composition ratio than the indium composition ratio of the first well layer, and the first intermediate layer is less in thickness than the second well layer.

(101) [Item 2] The light emitting element according to Item 1 above, wherein the second well layers are fewer in number than the first well layers.

(102) [Item 3] The light emitting element according to Item 1 or 2 above, wherein the second n-side semiconductor layer further includes a superlattice layer, the superlattice layer including a plurality of first layers containing indium and a second layer positioned between two adjacent first layers among the first layers, and the first layers having a lower indium composition ratio than the indium composition ratio of the first intermediate layer.

(103) [Item 4] The light emitting element according to any of Items 1 to 3 above, wherein the quantity of second well layers is one.

(104) [Item 5] The light emitting element according to any of Items 1 to 4 above, wherein the first intermediate layer has an indium composition ratio of 8% to 12%.

(105) [Item 6] The light emitting element according to any of Items 1 to 5 above, wherein the first well layer has an indium composition ratio of 12% to 18%, and the second well layer has an indium composition ratio of 20% to 25%.

(106) [Item 7] The light emitting element according to any of Items 1 to 6 above, wherein the second well layer is less in thickness than the first well layer.

(107) [Item 8] The light emitting element according to any of Items 1 to 7 above, wherein the first intermediate layer is 1 nm to 2 nm in thickness.

(108) [Item 9] The light emitting element according to any of Items 1 to 8 above, wherein the first well layer, the second well layer, and the first intermediate layer are formed of indium gallium nitride.

(109) [Item 10] The light emitting element according to any of Items 1 to 9 above further comprising successively a third n-side semiconductor layer in contact with the second p-side semiconductor layer, a third active layer having a third well layer containing indium, and a third p-side semiconductor layer, each being formed of a nitride semiconductor, wherein the third active layer has a second intermediate layer positioned closer to the second active layer than the third well layer is and containing indium, the second well layer has a lower indium composition ratio than the indium composition ratio of the third well layer, the second intermediate layer has a lower indium composition ratio than the indium composition ratio of the second well layer, and the second intermediate layer is less in thickness than the third well layer.

(110) [Item 11] The light emitting element according to Item 10 above, wherein the third well layers are fewer in number than the first well layers.

(111) [Item 12] The light emitting element according to Item 10 or 11 above, wherein the third well layer has an indium composition ratio of 25% to 40%.

(112) [Item 13] The light emitting element according to any of Items 10 to 12 above, wherein the second intermediate layer is 1 nm to 2 nm in thickness.

## Claims

1. A light emitting element comprising: a first n-side semiconductor layer; a first active layer located above the first n-side semiconductor layer and comprising a first well layer containing indium; a first p-side semiconductor layer located above the first active layer; a second n-side semiconductor layer located above and in contact with the first p-side semiconductor layer; a second active layer located above the second n-side semiconductor layer, the second active layer comprising: a first intermediate layer located above the second n-side semiconductor layer and containing indium, and a second well layer located above the first intermediate layer and containing indium; and a second p-side semiconductor located above the second active layer, wherein: each of the first n-side semiconductor layer, the first active layer, the first p-side semiconductor layer, the second n-side semiconductor layer, the second active layer, and the second p-side semiconductor is formed of a nitride semiconductor, an indium composition ratio of the first well layer is less than an indium composition ratio of the second well layer, an indium composition ratio of the first intermediate layer is less than an indium composition ratio of the first well layer, and a thickness of the first intermediate layer is less than a thickness of the second well layer.
2. The light emitting element according to claim 1, wherein a quantity of the second well layers is less than a quantity of the first well layers.
3. The light emitting element according to claim 1, wherein: the second n-side semiconductor layer further comprises a superlattice layer, the superlattice layer comprises: a plurality of first layers containing indium, and a second layer positioned between two adjacent first layers among the first layers, and an indium composition ratio of the first layers is less than the indium composition ratio of the first intermediate layer.
4. The light emitting element according to claim 2, wherein the second active layer includes exactly one second well layer.
5. The light emitting element according to claim 1, wherein the indium composition ratio of the first intermediate layer is in a range of 8% to 12%.
6. The light emitting element according to claim 2, wherein the indium composition ratio of the first intermediate layer is in a range of 8% to 12%.
7. The light emitting element according to claim 3, wherein the indium composition ratio of the first intermediate layer is in a range of 8% to 12%.
8. The light emitting element according to claim 1, wherein: the indium composition ratio of the first well layer is in a range of 12% to 18%, and the indium composition ratio of the second well layer is in a range of 20% to 25%.
9. The light emitting element according to claim 2, wherein: the indium composition ratio of the first well layer is in a range of 12% to 18%, and the indium composition ratio of the second well layer is in a range of 20% to 25%.
10. The light emitting element according to claim 3, wherein: the indium composition ratio of the first well layer is in a range of 12% to 18%, and the indium composition ratio of the second well layer is in a range of 20% to 25%.
11. The light emitting element according to claim 1, wherein the thickness of the second well layer is less than a thickness of the first well layer.
12. The light emitting element according to claim 2, wherein the thickness of the second well layer is less than a thickness of the first well layer.
13. The light emitting element according to claim 3, wherein the thickness of the second well layer is less than a thickness of the first well layer.
14. The light emitting element according to claim 1, wherein the thickness of the first intermediate

layer is in a range of 1 nm to 2 nm.

15. The light emitting element according to claim 1, wherein the first well layer, the second well layer, and the first intermediate layer are formed of indium gallium nitride.

16. The light emitting element according to claim 1 further comprising: a third n-side semiconductor layer located above and in contact with the second p-side semiconductor layer, a third active layer located above the third n-side semiconductor layer, the third active layer comprising: a second intermediate layer located above the third n-side semiconductor layer, and a third well layer located above the second intermediate layer and containing indium, and a third p-side semiconductor layer located above the third active layer, wherein: each of the third n-side semiconductor layer, the third active layer, and the third p-side semiconductor layer is formed of a nitride semiconductor, the indium composition ratio of the second well layer is lower than an indium composition ratio of the third well layer, an indium composition ratio of the second intermediate layer is lower than the indium composition ratio of the second well layer, and a thickness of the second intermediate layer is less than a thickness of the third well layer.

17. The light emitting element according to claim 16, wherein a quantity of the third well layers is less than a quantity of the first well layers.

18. The light emitting element according to claim 16, wherein the indium composition ratio of the third well layer is in a range of 25% to 40%.

19. The light emitting element according to claim 17, wherein the indium composition ratio of the third well layer is in a range of 25% to 40%.

20. The light emitting element according to claim 16, wherein the thickness of the second intermediate layer is in a range of 1 nm to 2 nm.

21. The light emitting element according to claim 17, wherein the thickness of the second intermediate layer is in a range of 1 nm to 2 nm.

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